

Estimating Groundwater Recharge from Soil Moisture Sensors in Semi-Arid Catchments

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ABSTRACT

Groundwater recharge in semi-arid regions is episodic and difficult to measure directly. We installed profiles of capacitance soil moisture sensors at 18 sites across two catchments and estimated drainage below the root zone with a water-balance model calibrated against lysimeter data. Annual recharge ranged from 4 to 61 mm, with more than 70% occurring during three storms. Sensor-based estimates agreed with chloride mass-balance estimates within 15% at 14 sites. The method offers a low-cost way to monitor recharge at the scale needed for water allocation.

Keywords: groundwater recharge; soil moisture; water balance; semi-arid hydrology

1. INTRODUCTION

Water managers in dry regions allocate groundwater on the basis of recharge estimates that are often decades old. Direct measurement with lysimeters is expensive, while tracer methods integrate over long periods and cannot resolve the response to individual storms. Low-cost soil moisture sensors are now widely deployed for irrigation scheduling, which raises the question of whether the same networks can be used to estimate recharge ...

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